

AYUSH SACHAN

Embedded Systems & Software Developer

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Summary

Embedded Systems and Software Developer with hands-on experience in designing and building real-time embedded systems using Arduino, ESP32/ESP8266, and Raspberry Pi. Skilled in C/C++, PCB design, and hardware-software integration, with experience in developing robotics systems including self-balancing robots and drones. Strong understanding of sensors, motor control, and communication protocols, with a passion for embedded and robotics innovation.

Education

- **Madan Mohan Malaviya University of Technology, Gorakhpur** **2023 – Present**
Bachelor of Technology (B.Tech) in Electrical Engineering
- Relevant Coursework: Embedded Systems, Digital Electronics, Microprocessors, Control Systems
- **Morning Star senior secondary school Orai** **2022**
Class XII (Senior Secondary) *Percentage: 75%*
- **Morning Star senior secondary school Orai** **2020**
Class X (Secondary) *Percentage: 82%*

Skills

Programming Languages: C/C++, JavaScript

Embedded Systems: Arduino, ESP32, ESP8266, Raspberry Pi

Hardware & Design: PCB Design, Circuit Design, Sensors & Actuators

Communication Protocols: UART, I2C, SPI, WiFi, Bluetooth

Software Development: Node.js, Express.js, React.js, Flutter, REST APIs

Databases: MySQL, MongoDB

Tools: Git, GitHub, VS Code, Arduino IDE, KiCad, Linux.

Core Strengths: Embedded Programming, Debugging, System Integration, Problem Solving

Experience

- **Summer Intern** **June 2026 – July 2026**
Bharat Heavy Electricals Limited (BHEL)
 - Observed industrial automation systems and electrical equipment used in manufacturing.
 - Gained exposure to PLC/SCADA systems, electrical maintenance, and industrial safety practices.
 - Studied power systems, control panels, and industrial instrumentation.
 - Learned industrial workflows and technical documentation in a large-scale manufacturing environment.

Projects

- **Smart Helmet & Vehicle-Based Accident Detection and Safety System | [GitHub](#)**
 - Developed an intelligent safety system that prevents bike ignition if the rider is not wearing a helmet using sensor-based detection.
 - Extended the system to support car safety through seatbelt detection mechanisms.
 - Implemented real-time accident detection using accelerometer and impact sensors.
 - Integrated GPS and GSM modules to send emergency alerts with live location to family members.
 - Designed a scalable embedded architecture adaptable for both two-wheelers and four-wheelers.
 - Focused on real-time processing, reliability, and safety-critical system design.
- **Drone Development & Robotics Systems**
 - Designed and developed a quadcopter drone using brushless DC motors and Electronic Speed Controllers (ESCs) for efficient thrust generation.
 - Integrated microcontroller-based system for coordinating multiple motors to maintain flight stability and autonomous flight.
 - Performed testing and tuning to achieve balanced flight and responsive control.
- **Adaptive UPS System (Ongoing) | [GitHub](#)**

- Designing an adaptive Uninterruptible Power Supply (UPS) system to optimize power utilization and ensure efficient backup management.
- Implemented control logic and initial circuitry for automatic switching between mains power and battery backup.
- Developing intelligent load management and battery optimization mechanisms to improve system efficiency and reliability.
- Focused on real-time monitoring and adaptive response based on changing power conditions.
- **Self Balancing Bot** | [GitHub](#)
 - Developed a two-wheeled self-balancing robot using ESP32 and MPU9250 IMU sensor based on inverted pendulum principles.
 - Implemented a PID control algorithm to maintain real-time balance by continuously adjusting motor speed.
 - Tuned control parameters through testing and iteration to achieve stable balancing and quick recovery from disturbances.

Positions of Responsibility

- **Vice President and Former Circuitary Lead**
Robotics Club, Madan Mohan Malaviya University of Technology, Gorakhpur
 - Leading strategic planning, technical operations, and inter-domain coordination for robotics projects and competitions.
 - Managing teams across circuitry, programming, and mechanical domains to ensure smooth project execution and system integration.
 - Led embedded system design including microcontrollers, sensors, and control circuits.
 - Collaborated across teams for full system integration in robotics projects.

Awards & Certifications

- Qualified Phase 1 of Aerothon 2026, demonstrating practical skills in UAV design, embedded systems, and aerospace engineering.
- Organized Techsrijan 2026, contributing to event coordination and technical management.
- Workshop on Drones and IoT (2024): Hands-on experience with drone systems, sensors, and embedded technologies.
- Third Prize — Institution Innovation Council (IIC) for technical innovation.
- Active participant in robotics and embedded systems workshops.